

ThermoQ

User Manual

A worked example on Al–Cu–Li (COST)
Using the calculation files shipped in test/

Switch the interface with Help → Language. Help → User Manual opens this PDF in the language currently selected.

1. Scope

This manual follows one alloy, Al–Cu–Li, with the COST assessment, from Thermo-Calc and Pandat batch files through extraction and plotting. The figures are computed from the .exp / CSV files in test/; they are the same objects ThermoQ writes to Excel.

Menus: File, Import, Plot, Tools, Help. Calculate sits on the main window (single composition and composition-space batch). Language labels refresh after Help → Language.

Menu	Purpose
Import → Pandat to ThermoQ	Load P.xlsx, Ts.xlsx (required); P-S.xlsx, Ts-S.xlsx (optional).
Tools → Generate Thermo-calc Batch File	Build a .tcm from Template0 + loop body (%Element%, %T%).
Tools → Extract Thermo-calc Results	Melting Range, Miscibility Gap, T-zero, TriST Zone.
Tools → Generate / Extract Pandat	.pbf batch (T-zero, Gibbs) and P/Ts, T0.xlsx, TriST.
Plot	Phase surfaces, Qtrue, liquidus vectors, k, T-zero, miscibility gap.

2. Installation

— Python 3.8+ with tkinter. `pip install -r requirements.txt` then `python main.py`.

— TriST workbooks require matplotlib and scipy. Interactive HTML requires plotly.

3. Case: melting range (Thermo-Calc)

Folder for templates: test/Generate Thermo-calc Batch File-Melting Range/Al-Cu-Li/. Folder for results: test/Extract Thermo-calc Results-Melting Range/Al-Cu-Li_COST/exp/ (filenames such as Al0.050Cu0.010Li_np-T.exp).

3.1 Generate the .tcm

1. Tools → Generate Thermo-calc Batch File.
2. Template0 = template0.txt (database, system, baseline s-c, first make file).
3. Loop template = template.txt. Placeholders %Cu% and %Li% are replaced on the Min/Max/Step grid (mass fraction).
4. Write the merged .tcm (Alltcm.tcm is already in that folder) and run it in Thermo-Calc. Each composition produces one *_np-T.exp.

Template0 (baseline composition, excerpt):

```

go da
sw tcal9
d-sys al cu li
l-s c
get
go p-3
s-c t=500,p=101325,n=1
s-c w(cu)=0.00001
s-c w(li)=0.00001
l-c
c-e
s-a-v 1 t 300 1000 12.5
save_workspace Al0.00Cu0.00Li_np-T.POL y
step normal
post
s-d-a x t
s-d-a y np(liquid)
make file Al0.00Cu0.00Li_np-T.exp

```

Loop body (one block per grid point):

```

back
s-c w(cu)=%Cu%
s-c w(li)=%Li%
l-c
c-e
s-a-v 1 t 300 1000 12.5
save_workspace Al%Cu%Cu%Li%Li_np-T.POL y
step normal
post
s-d-a x t
s-d-a y np(liquid)
make file Al%Cu%Cu%Li%Li_np-T.exp

```

3.2 Extract liquidus and solidus

1. Tools → Extract Thermo-calc Results → Melting Range. Select the exp folder. Default filter *_np-T.exp.
2. The filename supplies w(Cu) and w(Li). Blocks between “\$ PLOTTED COLUMNS ARE : T and NP(LIQUID)” and BLOCKEND are read.
3. Liquidus is the lowest T with NP(LIQUID) = 1; solidus is the highest T with NP(LIQUID) = 0. Optional Template1 writes a .tcm for files listed as errors in Status.

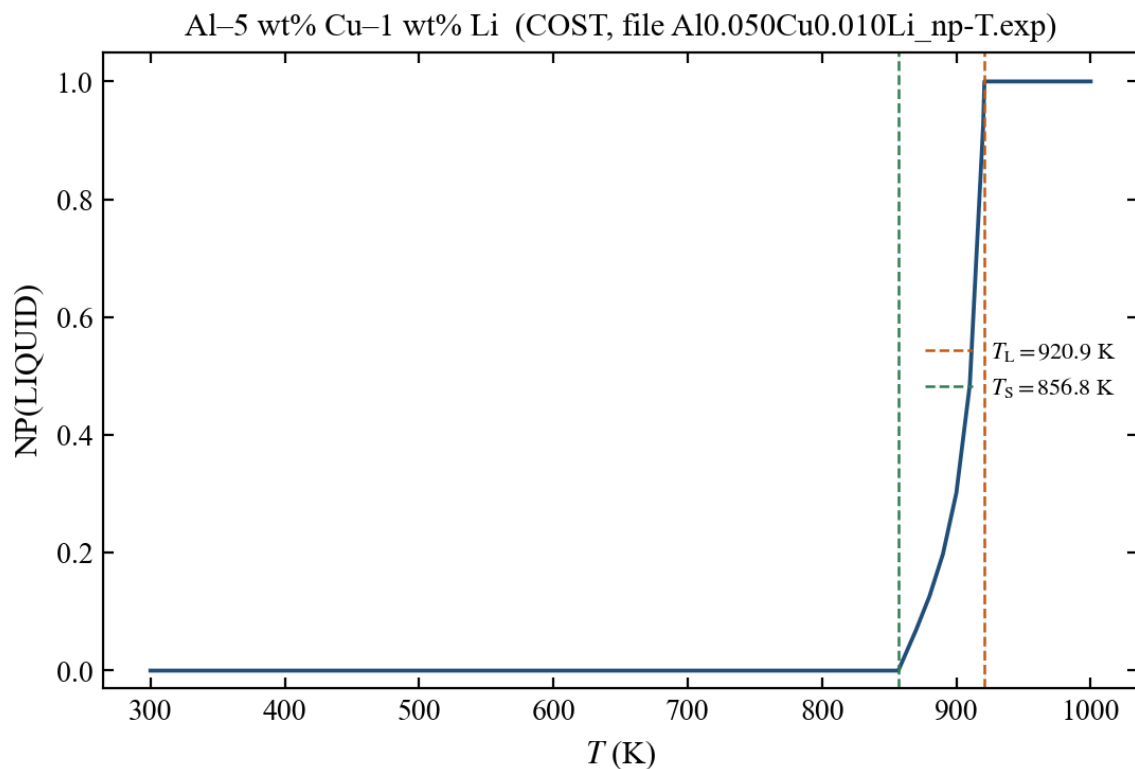


Fig. 1. $NP(LIQUID)$ versus temperature for Al-0.050Cu-0.010Li (COST). Vertical lines are the liquidus and solidus reported in the extract Excel.

Five rows from that extract (same algorithm as the Melting Range tab):

File	w(Cu)	w(Li)	TL / K	TS / K	ΔT / K
Al0.050Cu0.010Li_np-T.exp	0.050	0.010	920.9	856.8	64.1
Al0.100Cu0.020Li_np-T.exp	0.100	0.020	904.2	843.2	61.0
Al0.150Cu0.030Li_np-T.exp	0.150	0.030	880.7	836.8	43.9
Al0.180Cu0.005Li_np-T.exp	0.180	0.005	883.1	814.6	68.5

Extract Thermo-calc Results → Melting Range · Al-Cu-Li COST

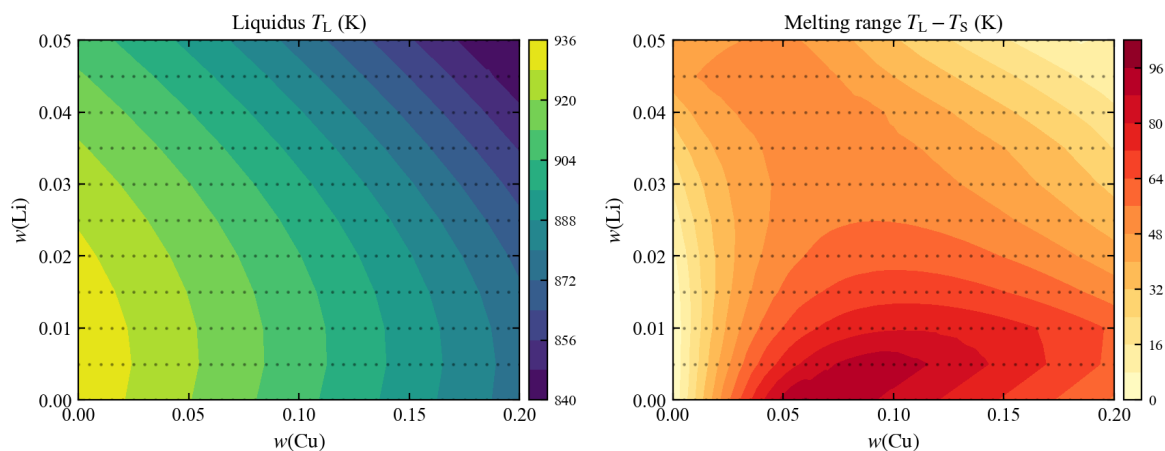


Fig. 2. Liquidus and melting range over the Al-Cu-Li COST grid in test/.../Al-Cu-Li_COST/exp. Dots are individual .exp files.
Plot → Phase Surfaces → Thermo-calc loads the Excel of this map.

4. Case: T-zero

Templates: test/Generate Thermo-calc Batch File-T0/Al-Cu-Li/. Results: test/Extract Thermo-calc Results-T0/Al-Cu-Li_COST/exp/AlCu-0.030Li_T0.exp, etc. The loop holds $w(\text{Li})$ fixed and scans $w(\text{Cu})$ from 0 to 0.2.

```
back
s-c w(Li)=%LI%
l-c
c-e
S-A-V 1
w(Cu)
0
0.2
0.005
save AlCu-%LI%Li_T0.POL y
step
t-z
FCC_A1
liquid
po
s-d-a x w(Cu)
s-d-a y t-k
make file AlCu-%LI%Li_T0.exp
```

1. Extract Thermo-calc Results → T-zero. Filename → $w(\text{Li})$; XTEXT $W(\text{Cu})$ → scan axis; plotted Y → T_0 (K).
2. Save `t_zero.xlsx`. Plot → Plot T-zero Surface (Thermo-Calc), set the isotherm interval.

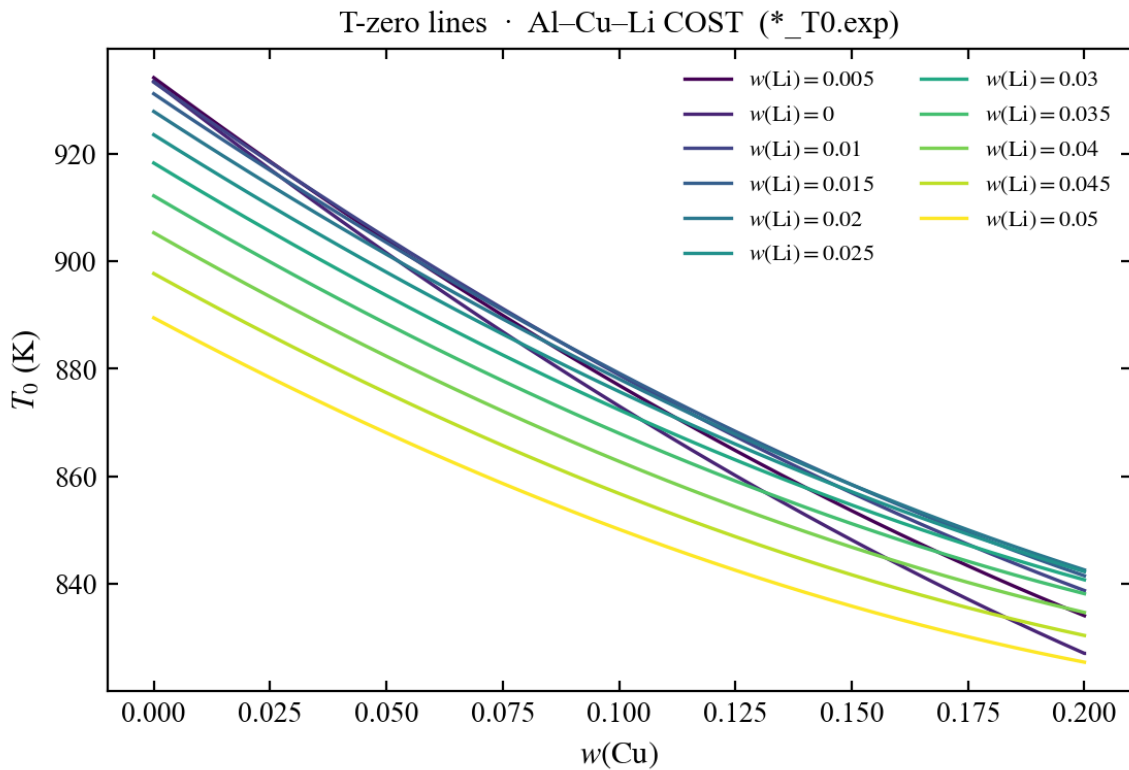


Fig. 3. $T_0(w(\text{Cu}))$ at fixed $w(\text{Li})$ from every *_T0.exp in the COST folder.

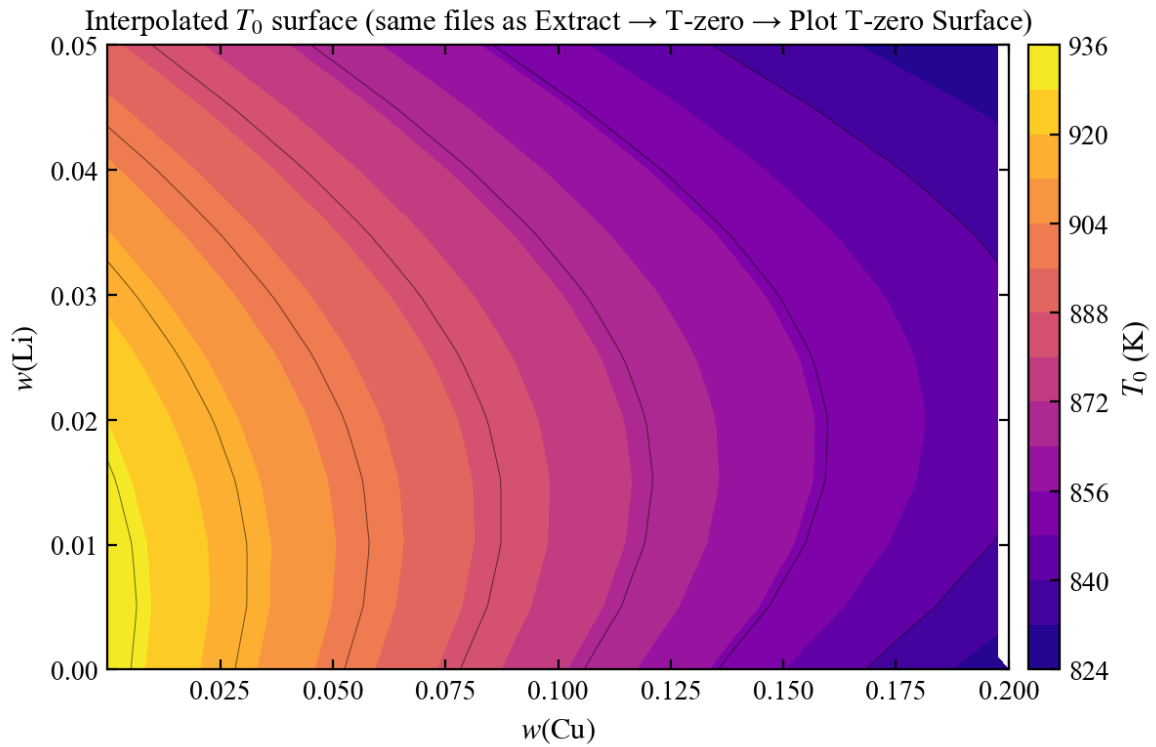


Fig. 4. Linear interpolation of those lines on the $w(\text{Cu})$ – $w(\text{Li})$ plane — the field drawn by Plot T-zero Surface.

5. Case: liquid miscibility gap

Files: test/Extract Thermo-calc Results-Miscibility Gap/Liquid/Al-Cu-Li_COST/exp/AlCuLi_800.exp (one temperature per file). Extract → Miscibility Gap parses T from the name, and XTEXT/YTEXT as mole % Cu and mole % Li.

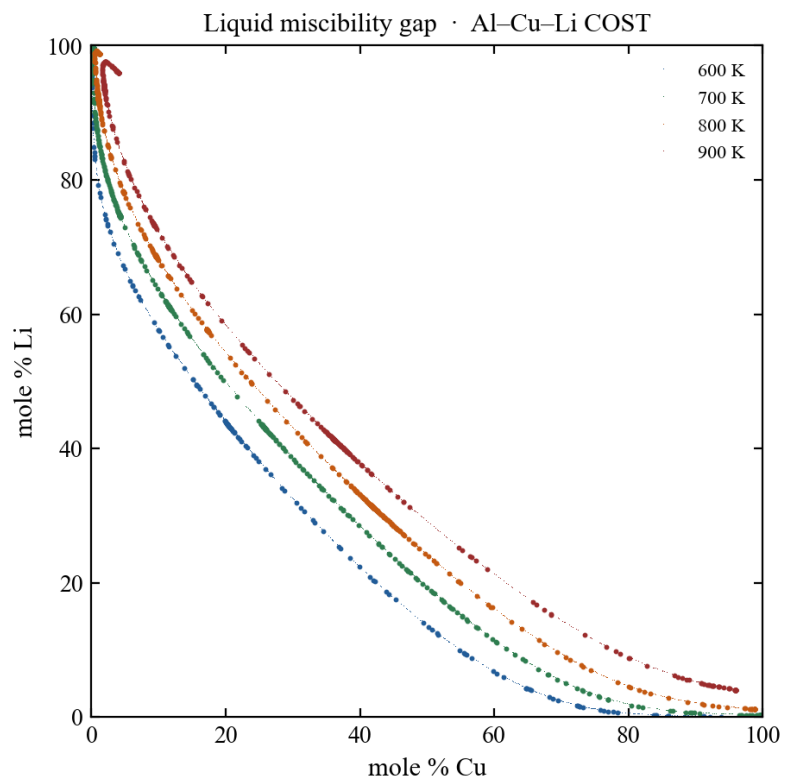


Fig. 5. Liquid miscibility-gap traces at 600–900 K from the COST .exp set. Plot → Plot Miscibility Gap reconstructs the same boundaries (with interpolation at a requested T).

6. Case: Gibbs energies and TriST Zone

Files: test/Extract Thermo-calc Results-Gibbs/Al-Cu-Li_COST/exp/, e.g. AlCu-0.050Li_650T_Gibbs.exp. Each file is one (w(Li), T); the scan axis is w(Cu). Sections are labelled “\$ PLOTTED COLUMNS ARE : W(CU) and GMR(PHASE)”.

1. Extract Thermo-calc Results → TriST Zone. Folder search is recursive; default filter `*_Gibbs\exp$`. Axes X/Y fill after Browse.
2. Build TriST workbook (T0_tie_1D, T0_lines, TriST_boundaries, TriST_mask, optional _trist_cube.npz). Grid N is typically 40–80.
3. Failed files go to Status; Template1 can emit one .tcm per error. Visualize with Plotly / 2D / 3D / GIF.

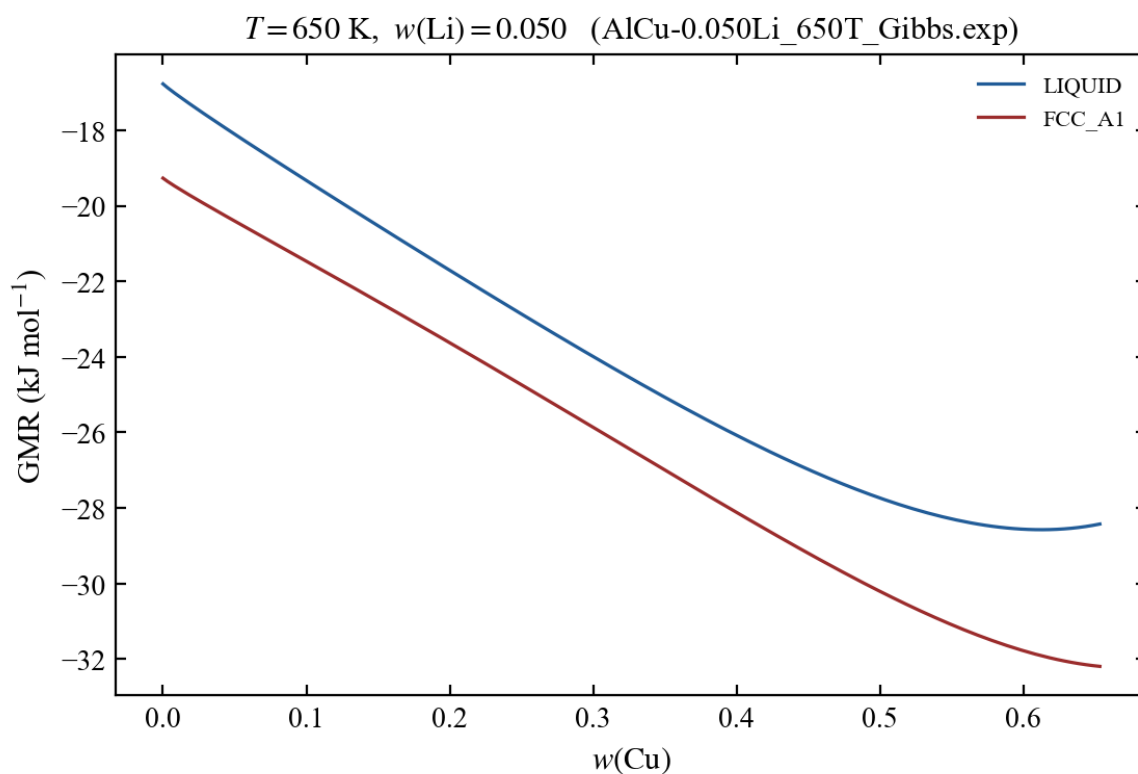


Fig. 6. Molar Gibbs energy of LIQUID and FCC_A1 versus $w(\text{Cu})$ at 650 K and $w(\text{Li}) = 0.050$, taken from one COST Gibbs .exp.

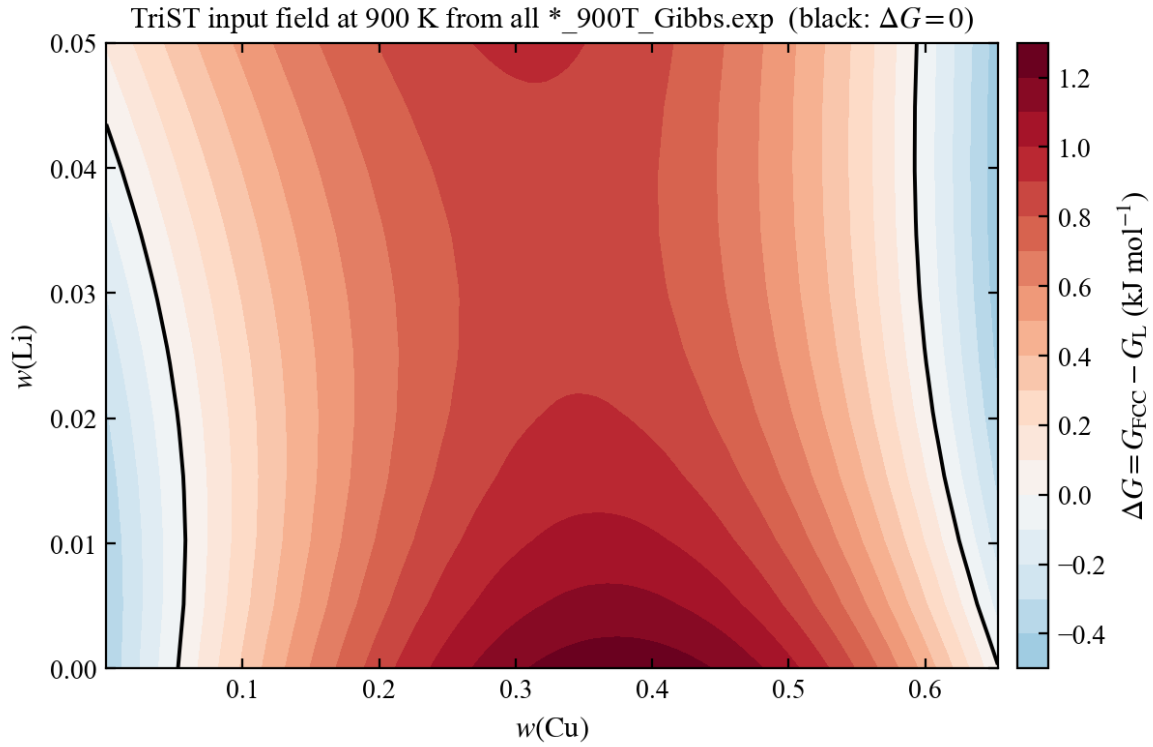


Fig. 7. $\Delta G = G(\text{FCC}) - G(\text{L})$ at 900 K assembled from all *_900T_Gibbs.exp in the same folder. The black contour is $\Delta G = 0$, i.e. the T_0 / TriST boundary at that temperature.

7. Case: Pandat Lever table (same alloy)

Batch templates: test/Generate Pandat Batch File-T0/Al-Cu-Li_COST/ and ...-Gibbs/. Solidification tables: test/Extract Pandat Results-P Ts (Lever Scheil)/Al-Cu-Li_COST/. Extract Pandat Results → P/Ts writes P.xlsx / Ts.xlsx / P-S.xlsx / Ts-S.xlsx, then Import to ThermoQ. Calculate uses Newton interpolation on those tables for Q_{true} , $Q/P/\beta$, ΔT and ΔT_s .

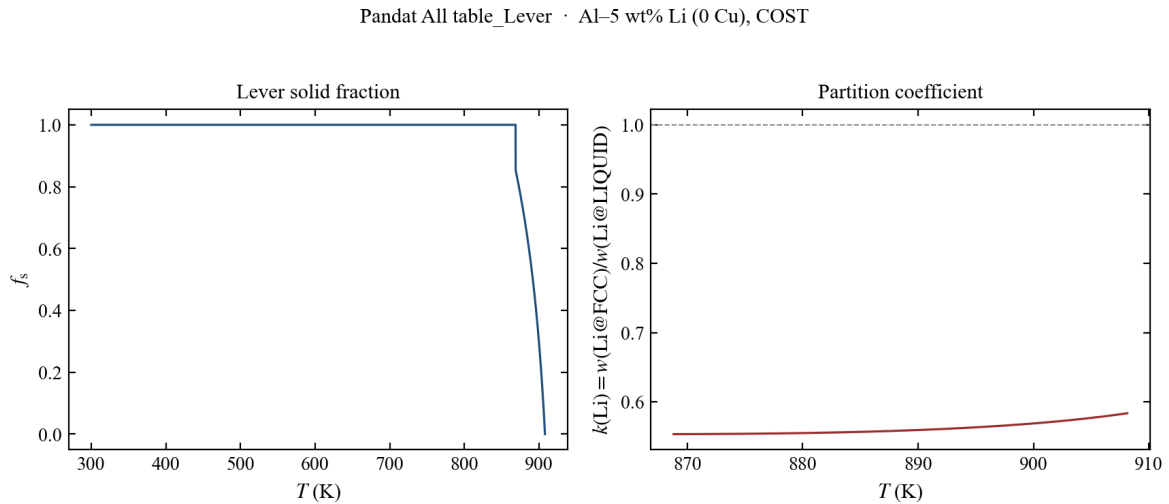


Fig. 8. One Lever CSV from that folder (Al-5 wt% Li, 0 Cu): solid fraction and $k(\text{Li})$. Plot → Solid-Liquid Partition Coefficients uses the same $k = w(*@solid)/w(*@LIQUID)$.

8. Plot tools after extract / import

- Phase Surfaces — Pandat P/Ts or the melting-range Excel of §3.
- Qtrue, liquidus vectors, partition k (liquidus / isotherm / isocomposition).
- T-zero Surface and Miscibility Gap — Excel from §4 and §5.
- Plot Labels: empty fields keep default axis names.

9. Where the other test/ trees fit

test/ folder	ThermoQ command
Generate Thermo-calc Batch File-{Melting Range, T0, Gibbs, Miscibility Gap}/	Generate Thermo-calc Batch File
Extract Thermo-calc Results-Melting Range/	Extract → Melting Range
Extract Thermo-calc Results-T0/	Extract → T-zero
Extract Thermo-calc Results-Gibbs/	Extract → TriST Zone
Extract Thermo-calc Results-Miscibility Gap/	Extract → Miscibility Gap
Generate Pandat Batch File-T0 / -Gibbs/	Generate Pandat Batch File
Extract Pandat Results-P Ts (Lever Scheil)/	Extract Pandat → P/Ts
Extract Pandat Results-T0/	Extract Pandat → T-zero
Extract Pandat Results-Gibbs/	Extract Pandat → TriST Zone

The same layout is repeated for Al–Cu–Si, Al–Mg–Li, Al–Mg–Si and Ti–Fe–Cu, with COST, TCAL9, He2009, Peisheng2011 or Bo2013 in the folder name. Help → Example opens the smaller Example/ tree.

10. Notes

- Close an extract workbook in Excel before overwriting it.
- Abnormal-point TCM files require a Template1 with %Element% and %T%, after Status has recorded failures.
- TriST needs scipy and matplotlib.